SAT Unit 3.2
Percentages

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Overview

Introduction to Percent

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Introduction to Percent

Definition: Percent means "per hundred." It represents a ratio of a number to 100.

$$\text{Percent} = \frac{\text{Part}}{\text{Whole}} \times 100$$

Example: 25 out of 100 is written as 25%.

Percent Change Formula

Formula:

$$\text{Percent Change} = \frac{\text{New Value} - \text{Original Value}}{\text{Original Value}} \times 100$$

Example: If a price increases from \$80 to \$90:

$$\frac{90 - 80}{80} \times 100 = 12.5\%$$

Key Concept: Positive percent change indicates an increase, negative indicates a decrease.

Percent Greater or Less Than

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Percent Greater or Less Than

	Translates to...	Example
$p\%$ of x	$\frac{p}{100} \times x$	15% of $x \Rightarrow \frac{15}{100} \times x = 0.15$
$p\%$ greater than x	$\left(1 + \frac{p}{100}\right) x$	15% greater than $x \Rightarrow \left(1 + \frac{15}{100}\right) x$
$p\%$ less than x	$\left(1 - \frac{p}{100}\right) x$	15% less than $x \Rightarrow \left(1 - \frac{15}{100}\right) x$

Example: If 200 is **x**% greater than 50, what is **x**?

$$50 \times \left(1 + \frac{x}{100}\right) = 200$$

Key Takeaway: If something is **50% greater than** a value, multiply by 1.5.

Question: Percent Relationship Between a, b, and c

The number a is **70% less** than the positive number b.
The number c is **60% greater** than a.
The number c is how many times b?

Answer Explanation: Percent Relationship Between a, b, and c

- Since a is 70% less than b, we express it as:

$$a = (1 - 0.7)b = 0.3b$$

- Since c is 60% greater than a, we express it as:

$$c = (1 + 0.6)a = 1.6a$$

- Substituting $a = 0.3b$:

$$c = 1.6(0.3b) = 0.48b$$

- Thus, the number c is **0.48** times b.

Correct Answer: $\boxed{0.48}$

Key Takeaways

- Percent represents "per hundred."
- Use the **percent change formula** to calculate increases and decreases.
- "Percent greater than" and "percent less than" refer to the original value.
- Use division to find the original value when given the increased price.

Question: Store Cost from Sale Price

The regular price of a shirt at a store is \$11.70.
The sale price of the shirt is **80% less** than the regular price, and the sale price is **30% greater** than the store's cost for the shirt.
What was the store's cost, in dollars, for the shirt?
(Disregard the \$ sign when entering your answer.)

Answer Explanation: Store Cost from Sale Price

- **Step 1: Find the sale price of the shirt**

$$\text{Sale Price} = (1 - 0.80) \times 11.70 = 0.20 \times 11.70 = 2.34$$

- **Step 2: Set up equation for the store's cost**

$$\text{Sale Price} = (1 + 0.30) \times \text{Cost}$$

$$2.34 = 1.30 \times \text{Cost}$$

- **Step 3: Solve for Cost**

$$\text{Cost} = \frac{2.34}{1.30} = 1.80$$

Correct Answer: $\boxed{1.80}$

Increased By vs. Decreased By

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"Increased By" vs. "Decreased By"

Definitions:

- **Increased by p%:** Multiply by $\left(1 + \frac{p}{100}\right)$
- **Decreased by p%:** Multiply by $\left(1 - \frac{p}{100}\right)$

Examples:

- 120 increased by 25%:

$$120 \times \left(1 + \frac{25}{100}\right) = 120 \times 1.25 = 150$$

- 150 decreased by 20%:

$$150 \times \left(1 - \frac{20}{100}\right) = 150 \times 0.8 = 120$$

Important Notes

Important Notes:

- **Be careful:** A 20% decrease followed by a 20% increase does **not** return to the original value!

$$100 \rightarrow 80 \rightarrow 96$$

- Always treat **"increased by" and "decreased by"** as multiplicative — not additive.
- If you are given a final value and percentage, **reverse the multiplier** to solve for the original.

Key Insight: Think in terms of **scaling**, not just adding or subtracting!

Question: Net Percentage Increase

The value of a collectible comic book increased by 167% from the end of **2011** to the end of **2012**, and then decreased by 16% from the end of **2012** to the end of **2013**.

What was the net percentage increase in the value of the collectible comic book from the end of **2011** to the end of **2013**?

A. 124.28%

B. 140.28%

C. 151.00%

D. 209.72%

Answer Explanation: Net Percentage Increase

Let the original value at the end of 2011 be 100 (for simplicity).

- After a 167% increase:

$$100 + 1.67 \times 100 = 267$$

- After a 16% decrease:

$$267 \times (1 - 0.16) = 267 \times 0.84 = 224.28$$

- Net increase from original 100:

$$224.28 - 100 = 124.28$$

  The net increase is 124.28%.

Correct Answer: $\boxed{A}$

Practice

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Question: Percent Greater Relationship

In 2010, the number of houses built in Town A was n percent greater than the number of houses built in Town B. If 700 houses were built in Town A in 2010, which expression represents the number of houses built in Town B in 2010?

A. $700\left(\frac{100+n}{100}\right)$

B. $700\left(\frac{100}{100+n}\right)$

C. $700\left(\frac{100-n}{100}\right)$

D. $700\left(\frac{100}{100-n}\right)$

Answer Explanation: Percent Greater Relationship

Let x be the number of houses in Town B.
Town A had $n\%$ more than Town B, so:

$$700 = x\left(\frac{100+n}{100}\right) \quad \Rightarrow \quad x = 700 \cdot \left(\frac{100}{100+n}\right)$$

Correct Answer: $\boxed{B}$

Question: Percent of c that is a

The sum of positive numbers a and b is 4.76 times the positive number c, and a is 124% of b. What percent of c is a?

A. 216.85%

B. 238.00%

C. 263.50%

D. 295.12%

Answer Explanation: Percent of c that is a

We are given:

$$a + b = 4.76c \quad \text{and} \quad a = 1.24b$$

Substitute a into the sum:

$$1.24b + b = 4.76c \Rightarrow 2.24b = 4.76c \Rightarrow b = \frac{4.76c}{2.24}$$

Now substitute b into $a = 1.24b$:

$$a = 1.24 \cdot \frac{4.76c}{2.24} = \frac{1.24 \cdot 4.76}{2.24}c = 2.635c$$

Convert to percent:

$$a = 263.5\% \text{ of } c$$

Correct Answer: $\boxed{C}$

Question: Percent of z that is x

The sum of positive numbers x and y is 3.5 times the positive number z, and x is 160% of y. What percent of z is x?

A. 212.00%

B. 233.33%

C. 256.00%

D. 280.00%

Answer Explanation: Percent of z that is x

Thank You!

We appreciate your time and focus.

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